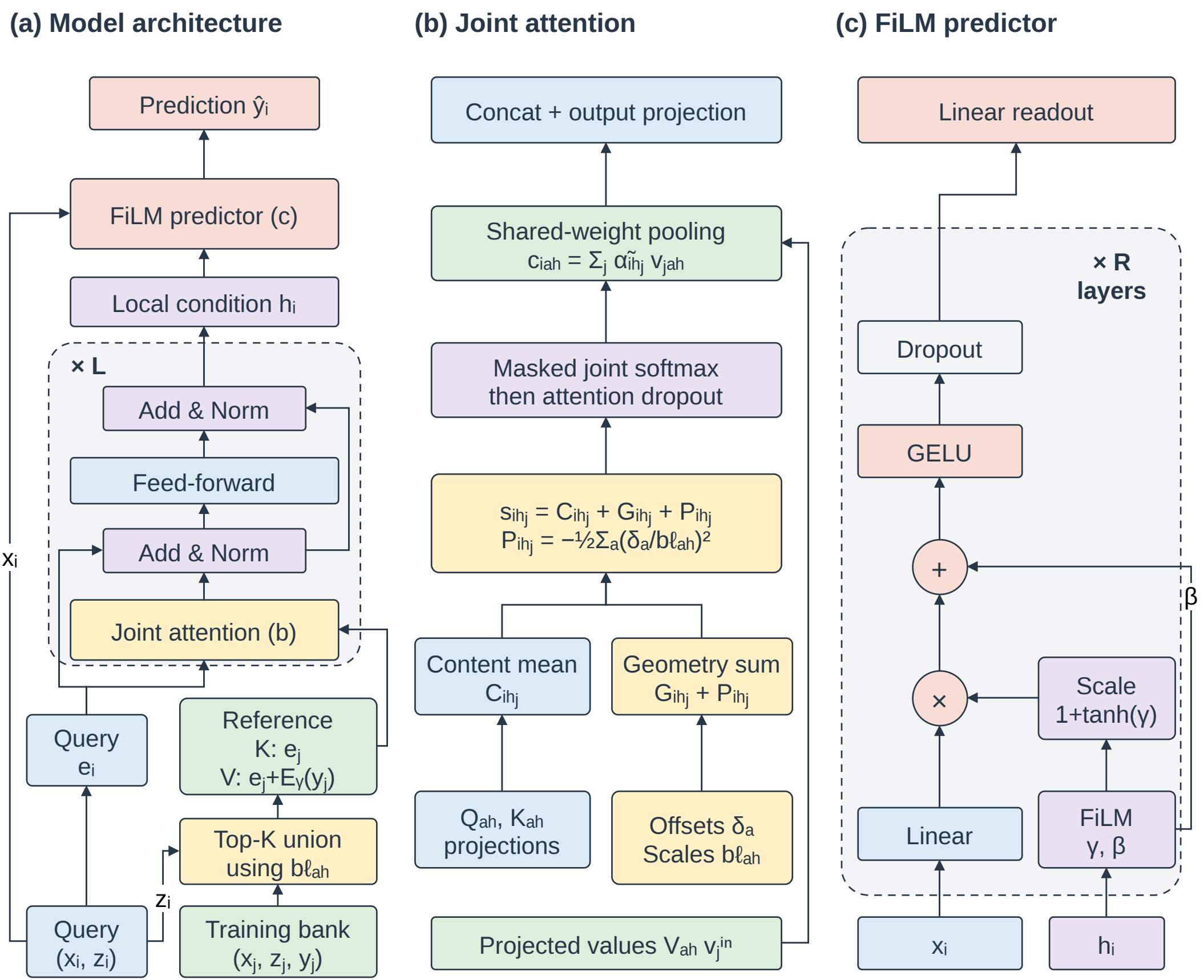
(a) Model architecture
Prediction ŷi
FiLM predictor (c)
Local condition hi
× L
Add & Norm
Feed-forward
Add & Norm
Joint attention (b)
xi
Query
ei
Reference
K: ej
V: ej+Eγ(yj)
Top-K union
using bℓah
zi
Query
(xi, zi)
Training bank
(xj, zj, yj)
(b) Joint attention
Concat + output projection
Shared-weight pooling
ciah = Σj α̃ihj vjah
Masked joint softmax
then attention dropout
sihj = Cihj + Gihj + Pihj
Pihj = −½Σa(δa/bℓah)²
Content mean
Cihj
Geometry sum
Gihj + Pihj
Qah, Kah
projections
Offsets δa
Scales bℓah
Projected values Vah vjin
(c) FiLM predictor
Linear readout
× R
layers
Dropout
GELU
+
β
×
Scale
1+tanh(γ)
Linear
FiLM
γ, β
xi
hi